

INTERIOR DOME LIGHTING – ILLUMINATION LEVEL

Advanced Attribute Performance and Benchmarking
January 13, 2026





Technical Protocol: Interior Dome Light - Illuminance Audit

Test Condition summary

1. Environmental & Vehicle Normalization

- **Zero-Check:** Ambient noise verified at **< 1 Lux**; noise sources covered if threshold exceeded.
- **Thermal Soak:** 5-minute active lamp stabilization to mitigate "**Thermal Drop**" (efficiency loss via heat).
- **EV State:** High-Voltage system in "**READY**" mode to provide a stable **13V–14V** Low-Voltage rail.

2. Geometric & Instrumental Standards

- **The Reading Plane:** Defined as a horizontal surface **500±20 mm above the Seat H-Point**.
- **Sensor used:** Testo 540 Luxmeter mounted on a **500mm-height box** to ensure height consistency. *Data counts as recorded: 5 seconds after mounting on a box and pressing HOLD button.*
- **Grid:** 4-Point Lap Plane (P1–P4)

3. Data Points

- **1-lamp activation** to audit component-level optical focus.
- **Full-cabin activation** to audit total atmospheric illuminance.





Summary : In cabin Illumination

Table: illuminance levels across 4 primary occupant positions (P1–P4) at a standardized 500mm "Reading Plane"

Point ID	Zone	Unit	LH900LX		VF6		VF8		BYD Atto 3	
			1 lamp on	All lamps on	1 lamp on	All lamps on	1 lamp on	All lamps on	1 lamp on	All lamps on
P1	Front Left	Lux	14	15	17	17	13	16	29	30
P2	Front Right		9	12	15	16	17	20	52	53
P3	Rear Left	Lux	121	127	3	3	40	41	114	120
P4	Rear Right		131	141	3	4	40	40	113	120



LH900LX: Raw Intensity vs. UX Friction

- **Luminous Output:** Recorded the highest intensity in the audit at **121–141 Lux** for the rear row.
- **Optical Misalignment:** The light beam is overly focused and targets the **occupant's knees** rather than the lap. When the captain's chair is reclined, the light fails to follow the occupant's head position, indicating a lack of proper diffusion.
- **UX Obstruction:** Activation buttons are "hidden" inside the lamp housing, forcing the passenger to search for them blindly. In a luxury flagship, this is a major "pain point"; controls should be migrated to the armrest or a dedicated HMI panel.

Point ID	Zone	Unit	LH900LX		VF6		VF8		BYD Atto 3	
			1 lamp on	All lamps on	1 lamp on	All lamps on	1 lamp on	All lamps on	1 lamp on	All lamps on
P1	Front Left	Lux	14	15	17	17	13	16	29	30
P2	Front Right		9	12	15	16	17	20	52	53
-										
P3	Rear Left	Lux	121	127	3	3	40	41	114	120
P4	Rear Right		131	141	3	4	40	40	113	120

Subjective Feeling	
FRONT ROW	NOT OK
REAR ROW	TOO BRIGHT



*Data counts as recorded: 5 seconds after mounting on a box and pressing HOLD button.

* Brightness on photos are different from reality. Photos taken with Flash - after HOLD function activated.



VF6

- Illuminance Deficit:** Objective data confirms a critical failure in the rear row, recording only **3–4 Lux**, which renders reading functionally impossible.
- Geometric Conflict:** For a tall occupant (186cm) wearing glasses, the light strikes the lenses directly, creating **Veiling Glare** that significantly reduces contrast and visibility on the reading plane.
- Interface Logic:** Features purely utilitarian physical buttons and lacks individual lamp control via the HMI, reinforcing its position as a budget-oriented model.

Subjective Feeling	
FRONT ROW	MARGINAL
REAR ROW	NOT OK

Point ID	Zone	Unit	LH900LX		VF6		VF8		BYD Atto 3	
			1 lamp on	All lamps on	1 lamp on	All lamps on	1 lamp on	All lamps on	1 lamp on	All lamps on
P1	Front Left	Lux	14	15	17	17	13	16	29	30
P2	Front Right		9	12	15	16	17	20	52	53
-										
P3	Rear Left	Lux	121	127	3	3	40	41	114	120
P4	Rear Right		131	141	3	4	40	40	113	120



*Data counts as recorded: 5 seconds after mounting on a box and pressing HOLD button.

* Brightness on photos are different from reality. Photos taken with Flash - after HOLD function activated.

VF8

- Component Sharing:** The front row utilizes the same utilitarian hardware and physical button logic as the VF6, showing a lack of differentiation despite its higher market status.
- Rear Row Success:** Significant improvement over the VF6 due to side-mounted lamps that achieve **40–41 Lux** at the lap plane, providing a comfortable reading environment. The "cool white" hue is utilitarian but acceptable given the consistent intensity.

Subjective Feeling	
FRONT ROW	MARGINAL
REAR ROW	MARGINAL

Point ID	Zone	Unit	LH900LX		VF6		VF8		BYD Atto 3	
			1 lamp on	All lamps on	1 lamp on	All lamps on	1 lamp on	All lamps on	1 lamp on	All lamps on
P1	Front Left	Lux	14	15	17	17	13	16	29	30
P2	Front Right		9	12	15	16	17	20	52	53
-										
P3	Rear Left	Lux	121	127	3	3	40	41	114	120
P4	Rear Right		131	141	3	4	40	40	113	120



*Data counts as recorded: 5 seconds after mounting on a box and pressing HOLD button.

* Brightness on photos are different from reality. Photos taken with Flash - after HOLD function activated.

BYD Atto 3

- Tech Ecosystem:** Lacks a physical master switch but compensates with a highly integrated **HMI and Voice Assistant interface**. Users can control specific zones or the entire cabin via voice commands.
- Spectral Perception:** Perceived Color Temperature is approximately **4000K**. This "Warm Color" avoids the harsh blue of 6000K which is considered detrimental for night vision.
- Positioning:** Rear lamps are strategically placed at the grab handles, ensuring the passenger's head does not shadow the reading plane.

Subjective Feeling	
FRONT ROW	Adequate
REAR ROW	Bright

Point ID	Zone	Unit	LH900LX		VF6		VF8		BYD Atto 3	
			1 lamp on	All lamps on	1 lamp on	All lamps on	1 lamp on	All lamps on	1 lamp on	All lamps on
P1	Front Left	Lux	14	15	17	17	13	16	29	30
P2	Front Right		9	12	15	16	17	20	52	53
-										
P3	Rear Left	Lux	121	127	3	3	40	41	114	120
P4	Rear Right		131	141	3	4	40	40	113	120



*Data counts as recorded: 5 seconds after mounting on a box and pressing HOLD button.

* Brightness on photos are different from reality. Photos taken with Flash - after HOLD function activated.



Thank You

